

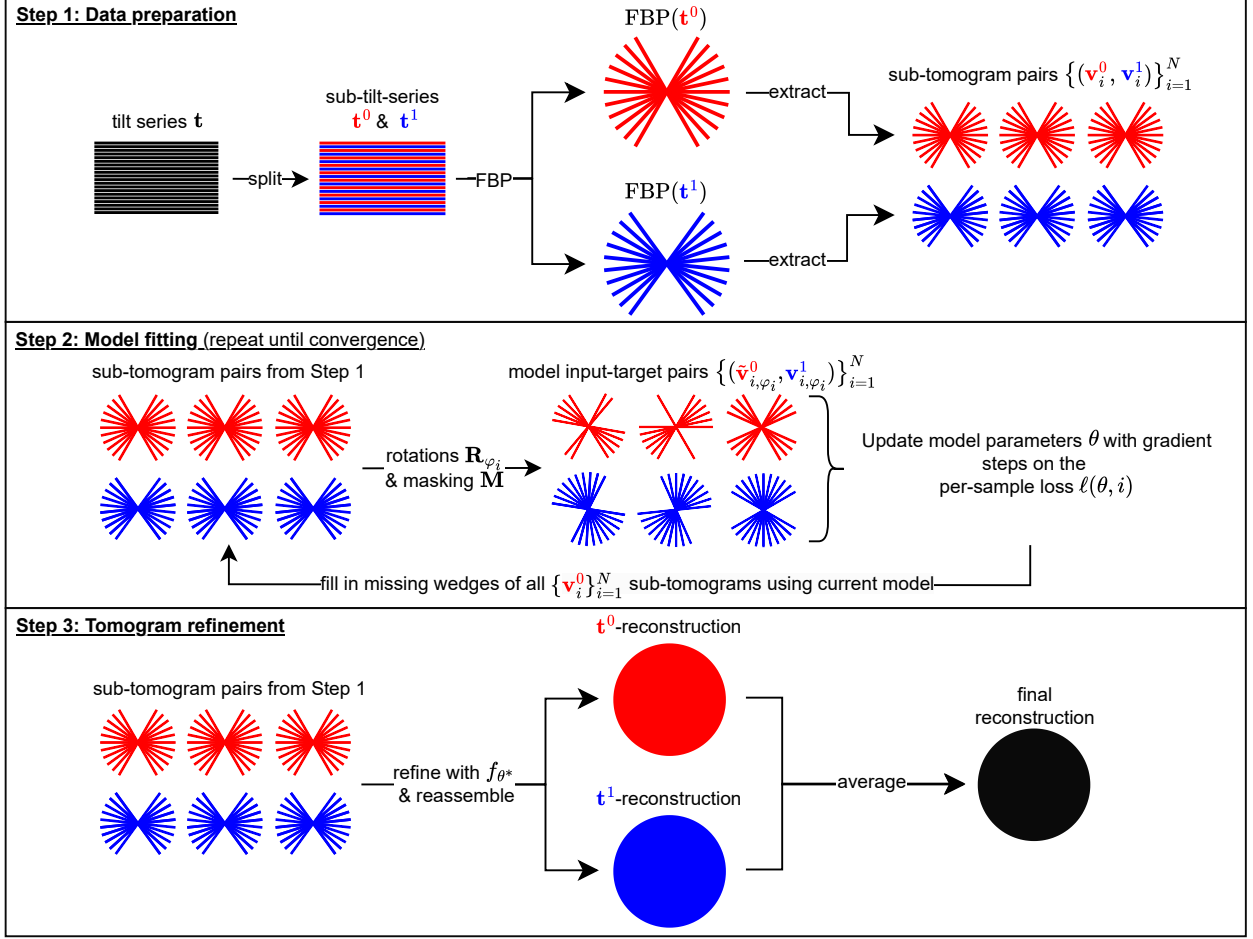


Figure 1: Illustration of DeepDeWedge. For simplicity, we show the 2D tilt series images as 1D Fourier slices and all 3D tomograms as 2D objects in the Fourier domain. Recall that tilt series images, tomograms, and sub-tomograms are objects in the image domain. The figure shows the splitting approach where the tilt series is split into even and odd projections.

propose fitting a randomly initialized network with a self-supervised loss for simultaneous denoising and missing wedge reconstruction. After fitting, the network is applied to the same data to estimate the tomogram. DeepDeWedge only uses the tilt series of the density we wish to reconstruct and no other training data. DeepDeWedge is most related to Noise2Noise-based denoising approaches and to IsoNet, both introduced above. We discuss the exact relations in Section 2.2 after describing our algorithm. DeepDeWedge performs on par with IsoNet on a pure missing wedge reconstruction problem and achieves state-of-the-art denoising performance on a pure denoising problem. Moreover, we find that the performance of DeepDeWedge for the joint denoising and wedge reconstruction problem is similar to the two-step approach of applying IsoNet to tomograms denoised with a state-of-the-art Noise2Noise-like denoiser. Moreover, DeepDeWedge is simpler and requires fewer hyperparameters to tune than the two-step approach.